

Project Demonstration

Date	15 March 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

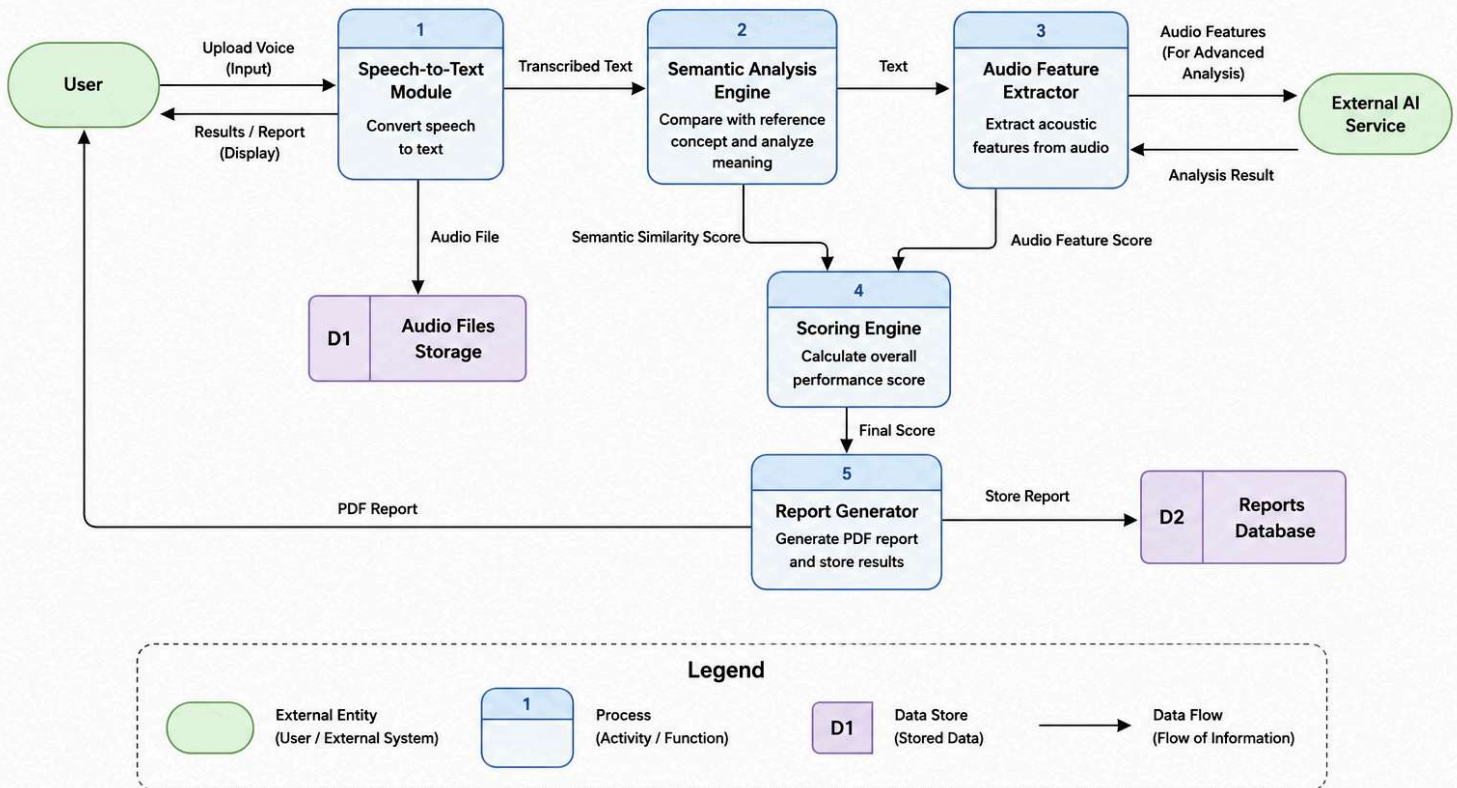
Project Demonstration Overview

This section demonstrates the complete workflow of the **Voice-Based Concept Understanding Analyzer**. The application accepts a student's voice explanation, converts speech into text, performs semantic analysis, evaluates conceptual understanding, and generates a detailed PDF report.

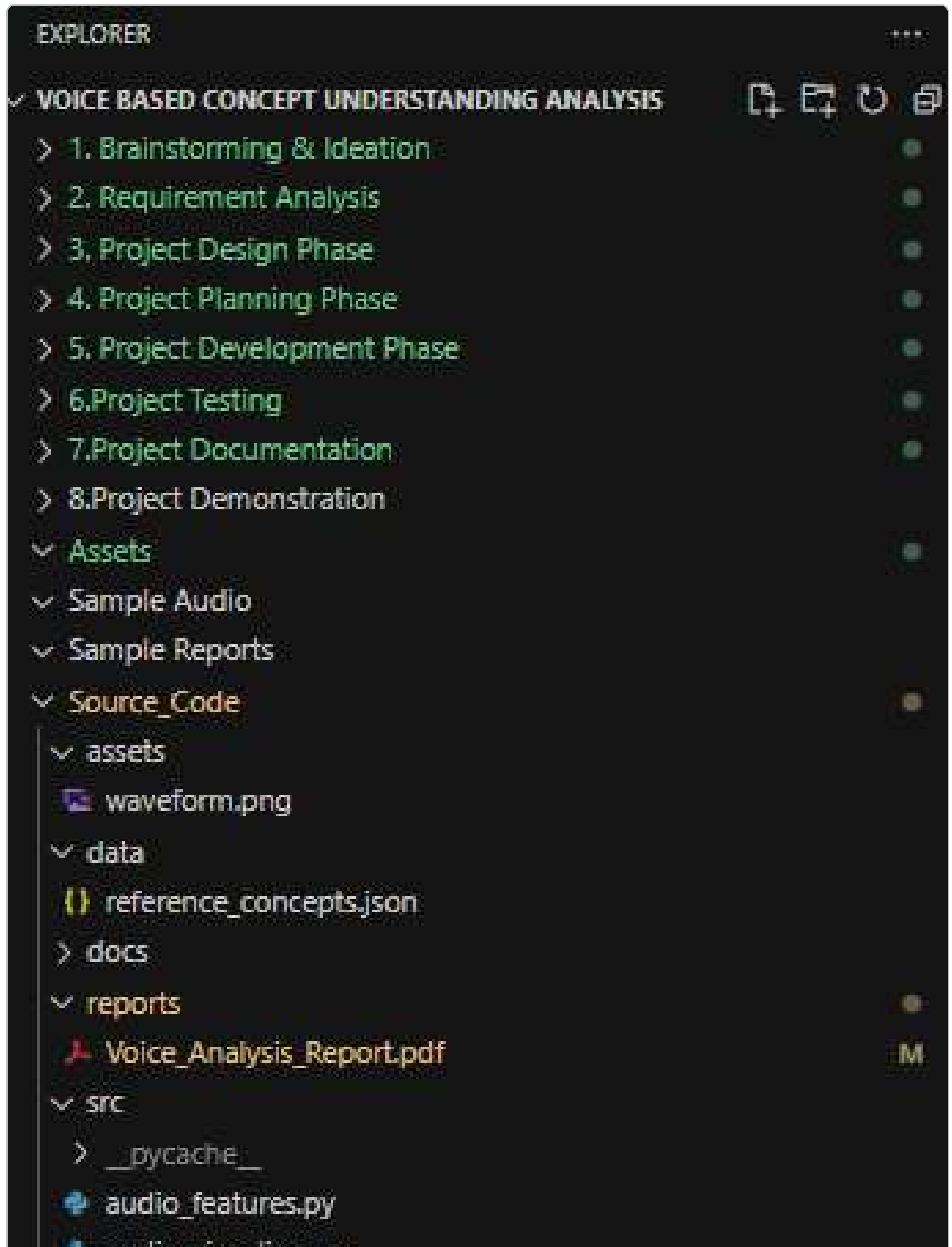
1. System Architecture

DATA FLOW DIAGRAM (LEVEL 0)

Voice-Based Concept Understanding Analyzer

**Figure 1: Data Flow Diagram**

2. Project Structure



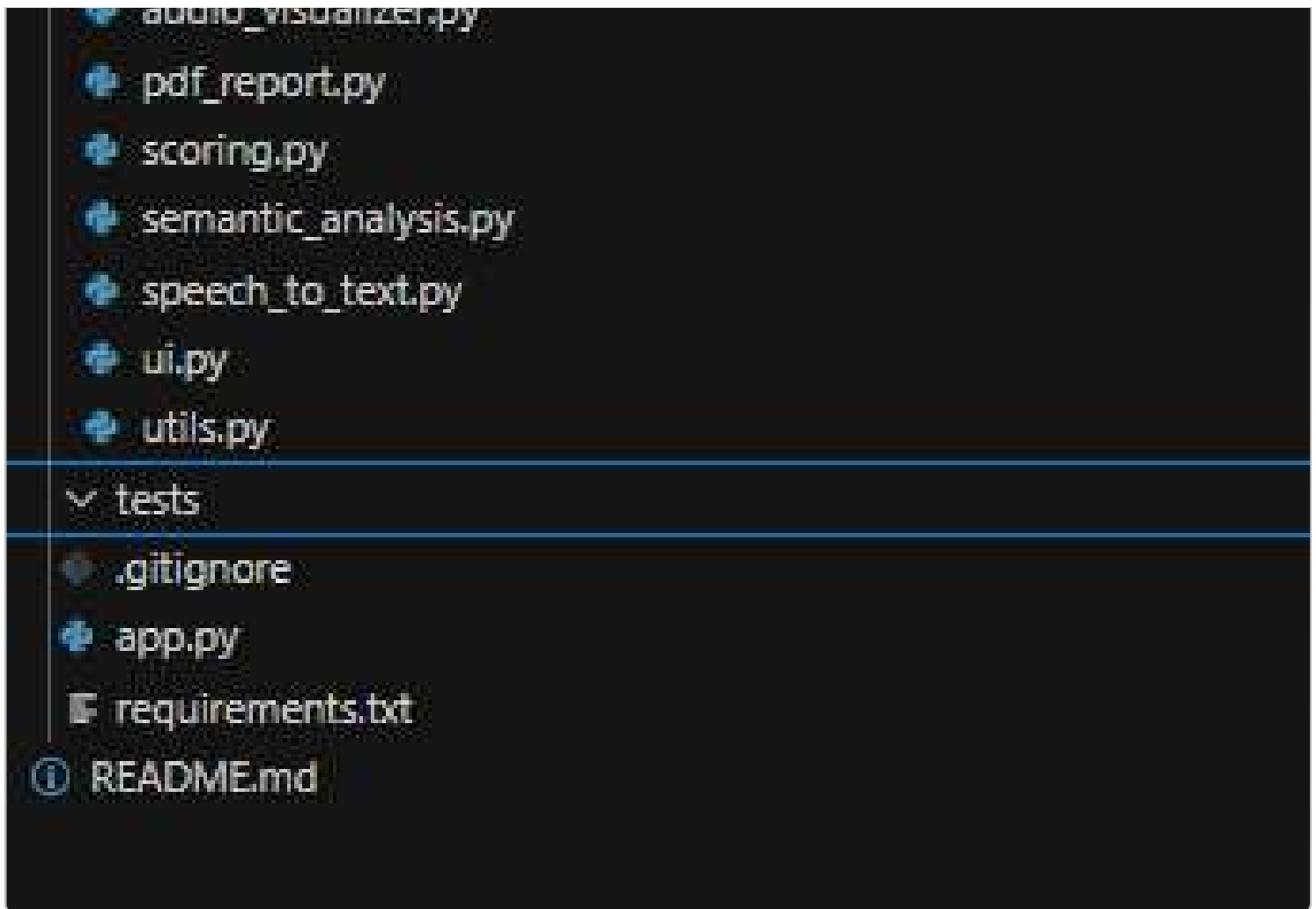


Figure 2: Project Folder Structure in VS Code

3. Running the Application

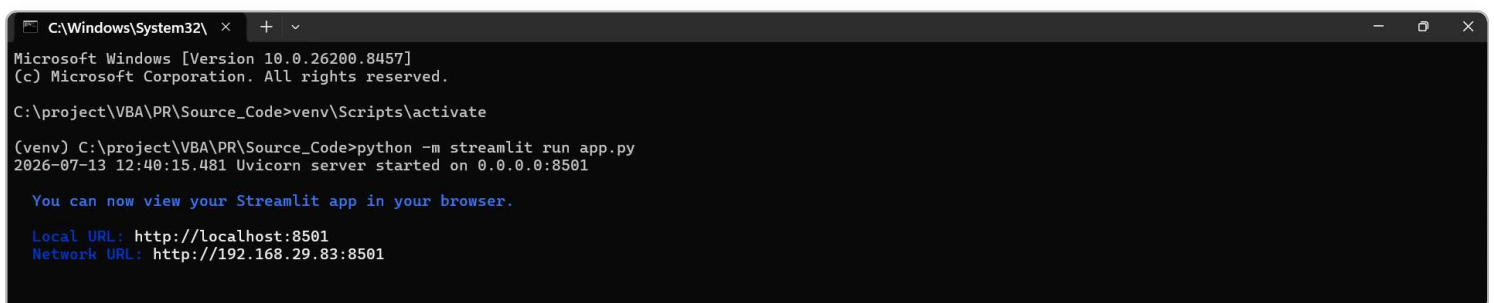


Figure 3: Streamlit Server Started Successfully

4. Streamlit Application Running

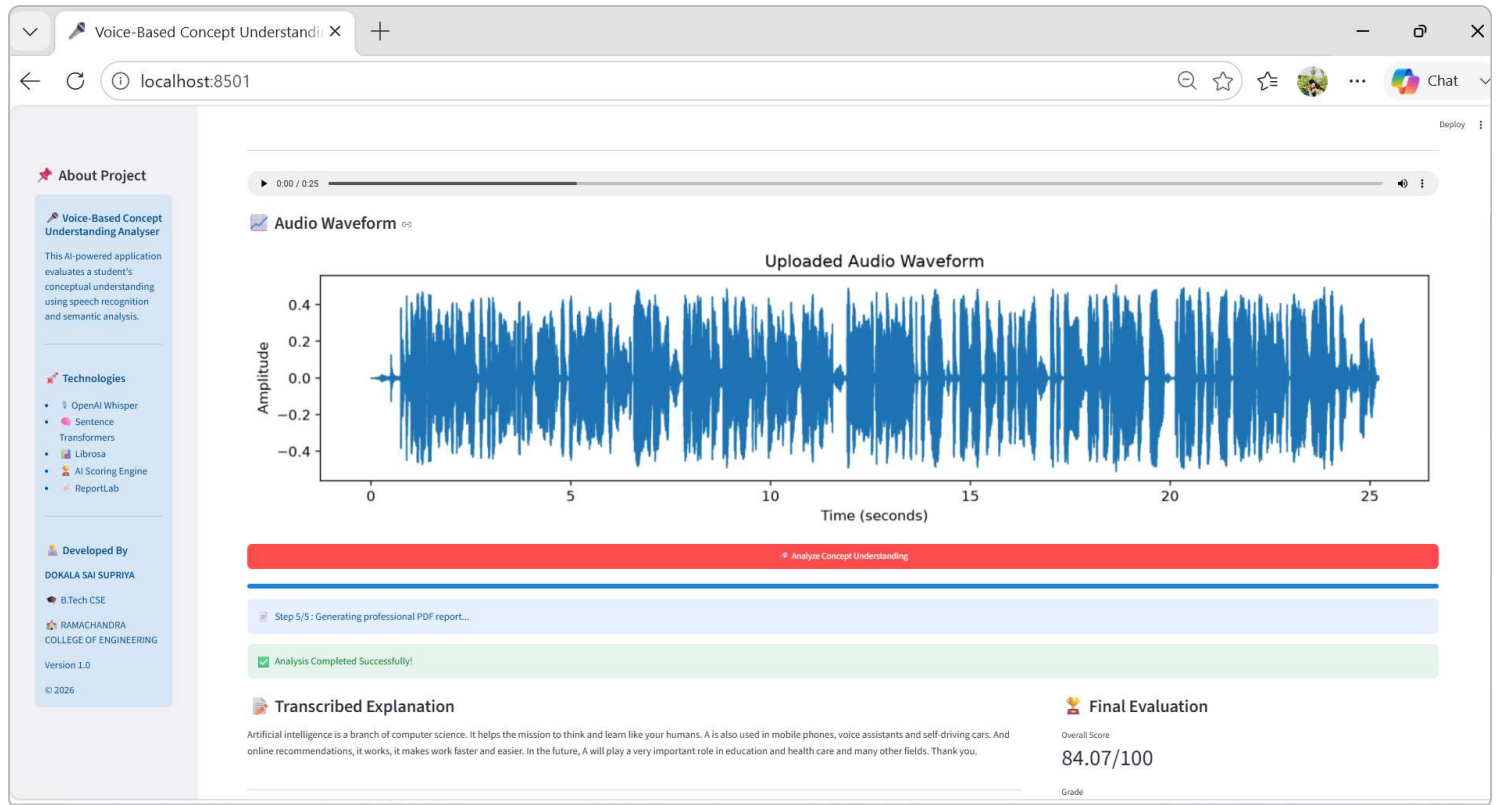


Figure 4: Application Running on Localhost

| 5. Application Workflow

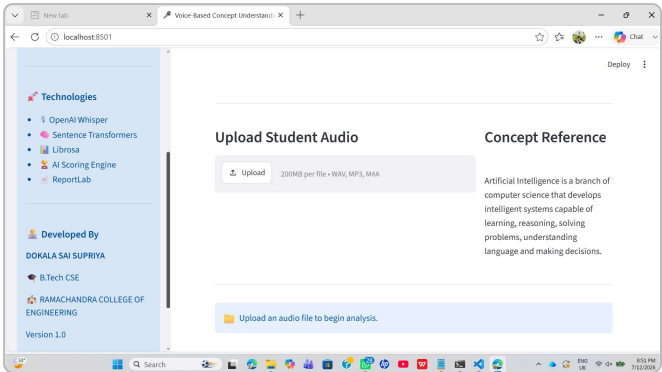


Figure 5: Home Page

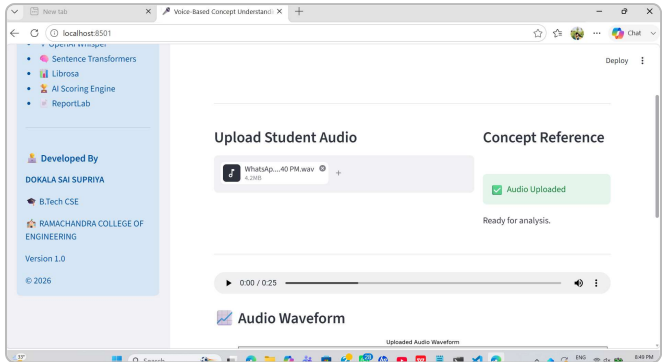


Figure 6: Audio Upload

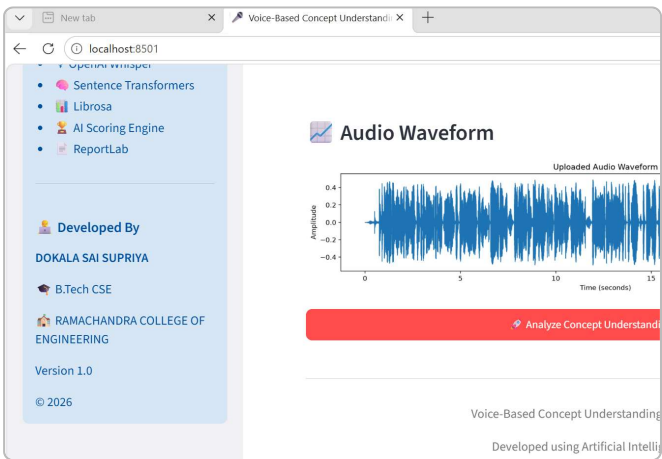


Figure 7: Audio Waveform

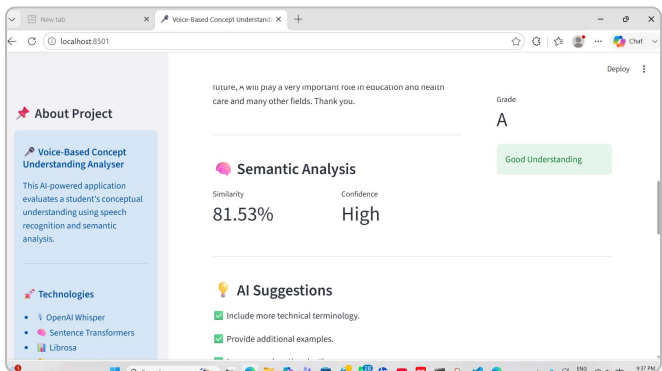


Figure 8: AI Evaluation Result

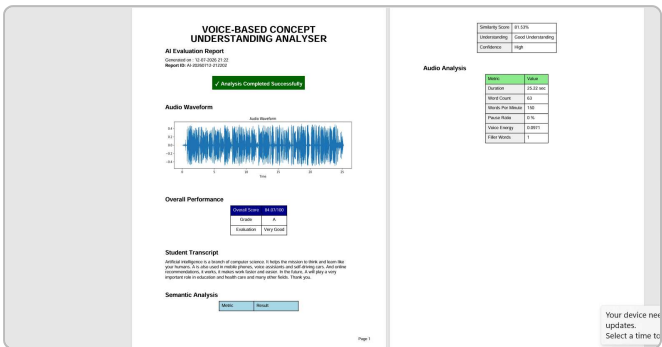


Figure 9: Generated PDF Report

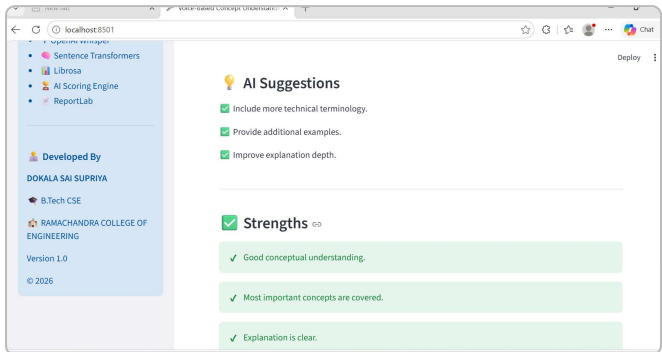


Figure 10: Strength & Semantic Analysis

6. Demonstration Summary

The application was successfully executed and tested. All major functionalities including audio upload, speech-to-text conversion, semantic similarity analysis, score generation, waveform visualization, dashboard display, and PDF report generation worked successfully during the demonstration.